

Alejandro Santin

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Projects:

- Data Search Website:

I developed the frontend of a website and connected it to the backend using a database through JavaScript. I used HTML, CSS, and JavaScript to link and manage the database. This experience helped me deepen my understanding of how websites function and how the frontend communicates with the backend.

- Encryption / Decryption program:

I developed a simple encryption/decryption program using functional programming principles in Haskell. This project provided me with a solid understanding of the Haskell programming language and allowed me to further refine my expertise in functional programming.

- OOP Game:

I successfully developed a two-player game using OOP, which also incorporated a GUI. Additionally, I implemented a simple AI that allowed the user to make calculated moves. This project helped me strengthen my OOP skills and provided valuable insight into implementing a more detailed and comprehensive testing phase for my code.

- AI Pattern Recognition Software

Using imported Python libraries, I developed an AI that, given an input (in the form of a spreadsheet), could recognise patterns and make calculated predictions based on trends in the data, estimating where the data would likely go for different inputs. This project enhanced my understanding of data processing and AI learning.

- ALU & ARM decode project

I developed an ALU for the 6502 CPU, which enhanced my understanding of the logic gates used in an ALU and how they work in unison to process arithmetic instructions and interact with other CPU components. Additionally, I created a program that decodes ARM instructions, identifying the instruction type, registers used and corresponding addressing mode by analysing the input bits, and presents the decoded instructions in plain English based on the ARM7TDMI manual.

- Client & Server Project

Utilising Python, I developed a simple client-server system that successfully communicated across a network. This project enhanced my understanding of networking principles and how communication occurs between clients and servers within a network.

- Mandelbrot in Assembly

I was tasked with implementing a code in assembly language to generate and output the Mandelbrot set for a given set of inputs. This project required me to dive deep into low-level programming, optimising the code for efficiency and ensuring that it accurately displayed the fractal for various input parameters. It was a challenging exercise that enhanced my understanding of both assembly language and the underlying mathematical principles behind fractals.

- A python maths quiz

Over two years, I designed, coded, debugged, and conducted research to create an interactive mathematics quiz in Python, aimed at helping younger students with their revision. I used Object-Oriented Programming (OOP) to structure the code efficiently, ensuring modularity and ease of updates. For managing user data, I implemented hashing to store and retrieve information quickly, and utilised linked lists for dynamic question handling, allowing for easy modification of the quiz content. Additionally, I integrated SQL for storing user data securely, applying encryption techniques to protect sensitive information. This project not only sharpened my coding and problem-solving skills but also deepened my understanding of building robust, secure, and scalable educational applications.

Skills Profile:

Throughout my projects and education, I have developed a broad set of technical and soft skills that have supported my progression in computer science. I have significantly strengthened my proficiency in Python, alongside a solid understanding of **OOP**, **SQL**, and data structures such as linked lists and hashing. My work has allowed me to explore a range of programming paradigms, including functional programming in Haskell and low-level development in ARM and **assembly** language, deepening my knowledge of CPU design, logic gates, and instruction decoding. I have also gained valuable experience in full-stack web development using **HTML**, **CSS**, **JavaScript**, and database integration. In addition, I have worked with other languages such as **C** and **Java**, which I applied in both academic and project settings. My exploration of artificial intelligence and data processing using Python libraries like Pandas and NumPy has further developed my ability to analyse data and recognise patterns. I have also built foundational knowledge in **networking** and client-server communication. These experiences have not only improved my technical expertise but also helped me develop key soft skills such as **time management**, **independent learning**, **research**, and **critical thinking**. I consistently take initiative and solve complex problems both independently and in **collaboration with others**. Additionally, my experience working as a lifeguard has helped me develop strong interpersonal skills that contribute to effective teamwork. The soft skills I developed include **communication**, **teamwork**, **decision-making**, and **situational awareness**.

Personal Statement:

I am a motivated Computer Science student currently seeking a placement opportunity where I can apply and further develop my technical skills in a professional environment. Through a range of academic and personal projects, I have gained valuable experience in programming, problem-solving, and understanding the broader field of computing. I am hardworking, well-organised, and able to perform effectively under pressure. I have strong communication skills, both in English and Spanish, which enables me to work confidently in diverse teams or customer-facing roles. I am also first aid and CPR certified and have completed training in the safe and effective use of an AED, reflecting my commitment to safety and responsibility in any setting. I am enthusiastic about contributing to your organisation and am eager to gain hands-on experience that will complement my academic learning and support my future career in technology.

Education:

University of Nottingham | Nottingham, England 08/2024-N/A
Studying for a Computer Science degree

Finchley Catholic High School | London, England 07/2017 – 05/2024
A Level 07/2022 – 05/2024
Subjects:

Computer Science -A
Maths-A
Physics-A

GCSE 07/ 2017 – 05/2022
Subjects:

Maths-9 Computer Science-9 Spanish-9
Physics-8 Biology-8 Chemistry-8
English Literature-7 English Language-7 French -7
Religious Education-7 Design and Technology-7

Past Experience:

- I have previously worked with my uncle at his mechanics I helped him carry boxes around and do any deliveries which he had alongside some manual labour for him. (06/2022-09/2022)
- I've also volunteered at charities before. (11/2022-12/2022)
- Additionally, I did waitering and worked behind the counter, both at my other uncle's restaurant for a short period of time. (06/2022-09/2022)
- I am currently employed as a lifeguard / recreational assistant (0 hours contract) where I gained communication skills and improved my confidence (04/2024-N/A)
- I have also been customer sales advisor where I learnt how to deal with customers in a professional manner (04/2024-08/2024)

Additional skills:

- Fluent in Spanish
- Advanced pseudocode knowledge
- NPQL certification
- AED certification
- SEQ Level 1 & 2